

Brief summary

Last time we have established that the sequence $\{k_{t+1}\}_{t=0}^{\infty}$ solves the (SP) problem $= F(k_t, k_{t+1})$ (denote)

$$(SP) \quad \max_{\{k_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t u'(f(k_t) - k_{t+1})$$

s.t. $0 \leq k_{t+1} \leq f(k_t)$, $t=1, 2, \dots$

k_0 is given $k_{t+1} \in F(k_t)$ (denote)

if and only if it solves

$$(EE): \quad u'(f(k_t) - k_{t+1}) = \beta f'(k_{t+1}) u'(f(k_{t+1}) - k_{t+2})$$

k_0 given $t=1, 2, \dots$

and

$$(TC) \quad \lim_{t \rightarrow \infty} \beta^t u'(f(k_t) - k_{t+1}) \cdot f'(k_t) \cdot k_t = 0$$

$\Rightarrow$ (SP) can be solved by finding a sequence $\{k_{t+1}\}$ that solves (EE) (e.g. "guess and verify" or
 lim. to a finite horizon probl.)
 and then verifying that (TC) is satisfied.

Solving the infinite horizon OGM

General strategy:

- find a solution of the (EE)
- check that (TC) is satisfied

What's a good "candidate" solution?

Example 1: The limit solution of the finite horizon model

Recall from last class:

$$u(c) = \log(c), \quad f(k) = Rk \Rightarrow$$

$$\Rightarrow S_t = \beta \cdot \frac{1 - \beta^{T-t}}{1 - \beta^{T-t+1}} \rightarrow \beta \text{ as } T \rightarrow \infty$$

$\Rightarrow$ Let's try $k_{t+1} = \beta R k_t$ as a possible solution

$$(EE): \frac{1}{Rk_t - \beta R k_t} = \beta R \frac{1}{Rk_{t+1} - \beta R k_{t+1}}$$

$$k_{t+1} = \beta R k_t \quad - \text{indeed holds}$$

$$(TC): \lim_{t \rightarrow \infty} \beta^t R \cdot \frac{1}{Rk_t - \beta R k_t} \cdot k_t = \lim_{t \rightarrow \infty} \beta^t \cdot \frac{1}{1 - \beta} = 0$$

also holds $\Rightarrow$

$\Rightarrow k_{t+1} = \beta R k_t$ is the optimal path!

Example 2: We can try to guess the functional form of the solution by looking at the (EE):

$$u(c) = \log(c), \quad f(k) = A k^\alpha$$

$$(EE): \frac{1}{A k_t^\alpha - k_{t+1}} = \beta A \alpha k_{t+1}^{\alpha-1} \cdot \frac{1}{A k_{t+1}^\alpha - k_{t+2}}$$

$$\text{Guess: } k_{t+1} = s \cdot A k_t^\alpha$$

$\uparrow$
i.e. guess that the savings rate is constant

$$\frac{1}{(1-s) A k_t^\alpha} = \beta \alpha \frac{A k_{t+1}^{\alpha-1}}{k_{t+1}} \cdot \frac{1}{(1-s) A k_{t+1}^\alpha}$$

$$k_{t+1} = \alpha$$

$$\frac{k_{t+1}}{A k_t^2} = \beta \alpha$$

$$\underbrace{\frac{k_{t+1}}{A k_t^2}}_{=s} \text{ (by guess)} \Rightarrow \boxed{s = \beta \alpha}$$

Check the (TC):

$$\lim_{t \rightarrow \infty} \beta^t A \alpha k_t^{\alpha-1} \cdot \frac{1}{(1-s) A k_t^2} \cdot k_t =$$

$$= \lim_{t \rightarrow \infty} \beta^t A \alpha \frac{k_t^{\alpha}}{(1-s) A k_t^2} = 0 \quad \text{holds!}$$

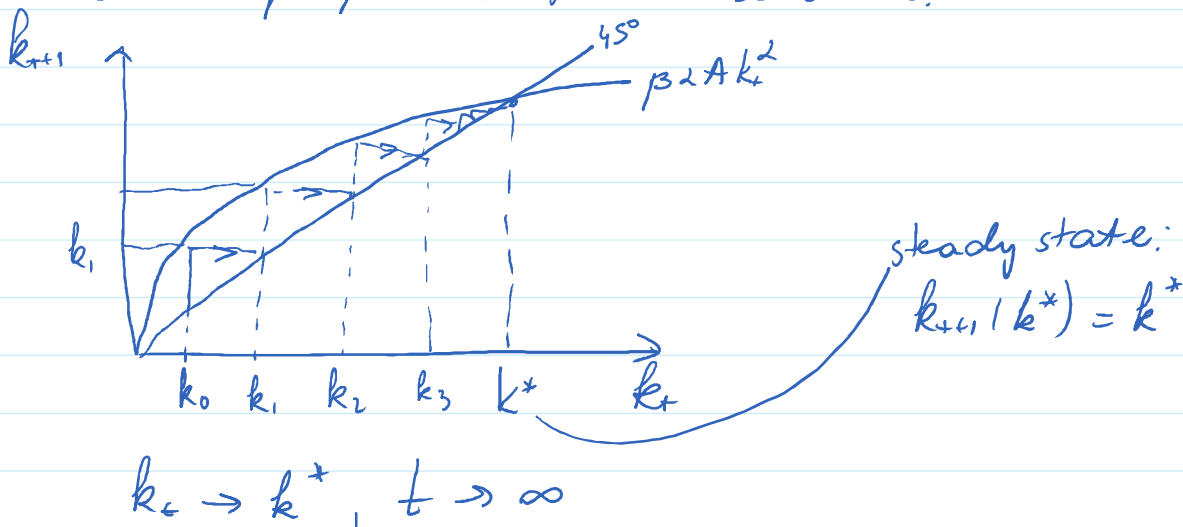
$$\Rightarrow \boxed{k_{t+1} = \beta \cdot \alpha \cdot A k_t^{\alpha}, \quad t \geq 0}$$

is the optimal $\{k_t^*\}_{t=0}^{\infty}$
 k_0 given

This approach is called "Guess and verify".

Intuition: $\beta \uparrow \Rightarrow$ more patient, value future consumption more $\Rightarrow$
 $\Rightarrow$ invest more

Some properties of this solution:



Steady State

Sometimes we are interested only in finding the steady-state (analyzing long-run effects of changes in economic environments).

Steady state: all variables are constant over time.
($k_t = k^*$, $\forall t$)

This is easy:

1) Just solve the (EE):

$$\cancel{u'(f(k^*) - k^*)} = \beta f'(k^*) \cdot \cancel{u'(f(k^*) - k^*)}$$
$$\Rightarrow \beta f'(k^*) = 1$$

(In the steady state the marginal product of capital $f'(k^*) = \frac{1}{\beta}$)

2) (TC) holds by default:

$$\lim_{T \rightarrow \infty} \underbrace{\beta^T f'(k^*) \cdot u'(f(k^*) - k^*)}_{\text{const}} \cdot k^* = 0$$

Therefore, any path $\{k_{t+1}\}_{t=0}^{\infty}$ that converges to the steady state satisfies (TC)

The question though is whether (under which conditions) $\{k_{t+1}\}_{t=0}^{\infty}$ converges to a steady state...

Open questions

Thus, we know

- how to find the Steady state in the optimal growth ^{model}
- how to solve it analytically for some $u(\cdot)$ & $f(\cdot)$.

But what if analytical solution does not exist?

- We could still try to find the steady state numerically
- Use the numerical methods to find the path of convergence to the steady state (if the economy converges to a steady state from k_0)...

Can we do more? (establish analytical properties of $\{k_{t+1}\}_{t=0}^{\infty}$?
stability of the steady state?)

New tools needed....

It turns out that a lot more can be achieved if we reformulate (SP) recursively and use Dynamic Programming Theory to characterize the solution!

Dynamic Programming

Denote by $V(k_0)$ the indirect utility function obtained after (SP) is solved:

$$(*) \quad V(k_0) = \max_{\{k_{t+1}\}_{t=0}^{\infty}} \left\{ \sum_{t=0}^{\infty} \beta^t F(k_t, k_{t+1}) \mid \text{s.t. } k_{t+1} \in \Gamma(k_t), t=0,1,\dots \right\}$$

$k_0 \text{ is given}$

Since $\max_{x,y} f(x,y) = \max_x \{ \max_y f(x,y) \}$, we can

rewrite (*) as :

$$V(k_0) = \max_{k_1} \left\{ \max_{\{k_2, k_3, \dots\}} \left\{ \sum_{t=0}^{\infty} \beta^t F(k_t, k_{t+1}) \mid \text{s.t. } k_{t+1} \in \Gamma(k_t), t=1,2,\dots \right\} \right\}$$

$k_1 \text{ is given}$

$\text{s.t. } k_1 \in \Gamma(k_0), k_0 \text{ is given}$

$= F(k_0, k_1) + \beta \cdot \sum_{t=0}^{\infty} \beta^t F(k_{t+1}, k_{t+2})$
 does not depend on $\{k_2, k_3, \dots\}$.

$= V(k_1) !!!$

$\Rightarrow$

$$V(k_0) = \max_{k_1} \left\{ F(k_0, k_1) + \beta \max_{\{k_2, k_3, \dots\}} \left\{ \sum_{t=0}^{\infty} \beta^t F(k_{t+1}, k_{t+2}) \mid \text{s.t. } k_{t+2} \in \Gamma(k_{t+1}), t=0,1,\dots \right\} \right\}$$

$k_1 \text{ is given}$

$\text{s.t. } k_1 \in \Gamma(k_0), k_0 \text{ is given}$

The original decision problem satisfies the Principle of Optimality: given the choice of k_1 , the remaining $\{k_2, k_3, \dots\}$ are also the outcome of the maximization problem (which is independent of k_0).

$$V(k_0) = \max_{k_1} \{ F(k_0, k_1) + \beta V(k_1) \}$$

$\text{s.t. } k_1 \in \Gamma(k_0)$

Here $V(k_0)$ - the value which the agent obtains if starts with k_0 and chooses $\{k_1, k_2, k_3, \dots\}$

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$V(k_1)$ - the value — " — it starts with k_1 and chooses $\{k_2, k_3, k_4, \dots\}$.

Note: we could similarly rewrite

$$V(k_1) = \max_{k_2} \{F(k_1, k_2) + \beta V(k_2)\} \\ \text{s.t. } k_2 \in \Gamma(k_1)$$

Or for period t :

$$V(k_t) = \max_{k_{t+1}} \{F(k_t, k_{t+1}) + \beta V(k_{t+1})\} \\ \text{s.t. } k_{t+1} \in \Gamma(k_t)$$

Intuitively, the agent makes an optimal decision in period t knowing that he will behave optimally from period $t+1$ onward (and hence will get indirect utility $V(k_{t+1})$)

In fact, we do not need the time index:
think in terms of "today" and "tomorrow" only:

(p)
$$V(k) = \max_{k'} \{F(k, k') + \beta V(k')\} \\ \text{s.t. } k' \in \Gamma(k)$$
 Bellman Equation!

dynamic programming problem

= Bellman equation

= "recursive problem"

Here: k - current period capital = state variable

k' - next period capital, chosen when k is known = choice variable
or
control variable.

Notice that if $V(k')$ were known, it would be

easy to solve for the optimal k' using Kuhn-Tucker method.

But we do not know what $V(k')$ is!!!

Since (RP) has $V(\cdot)$ in the LHS and RHS $\Rightarrow$
 $\Rightarrow$ it is an equation with respect to (unknown) $V(\cdot)$
 = FUNCTIONAL EQUATION

The solution to (RP) consists of:

1) $V(k)$ - VALUE FUNCTION

2) $g(k) = \arg \max_{k' \in \Gamma(k)} \{F(k, k') + \beta V(k')\}$ - POLICY FUNCTION

Obvious questions:

- Does (RP) have a solution?
- Is a solution to (RP) (if exists) unique?
- How to find it?

Note: The finite horizon optimal growth problem also satisfies the principle of optimality
 $\Rightarrow$ it can be also written down recursively:

$$\begin{aligned}
 V_0(k_0) &= \max_{k_1} \max_{\{k_2, k_3, \dots, k_{T+1}\}} \left\{ \underbrace{F(k_0, k_1) + \beta F(k_1, k_2) + \dots + \beta^T F(k_T, k_{T+1})}_{T \text{ terms}} \mid \begin{array}{l} k_{T+1} \in \Gamma(k_T) \\ k_2 \in \Gamma(k_1) \\ k_1 \in \Gamma(k_0) \end{array} \right\} \\
 &\quad \text{not impacted by } \{k_2, \dots, k_{T+1}\} \\
 &= \max_{k_1} \left\{ F(k_0, k_1) + \beta \max_{\{k_2, k_3, \dots, k_{T+1}\}} \left\{ \underbrace{F(k_1, k_2) + \dots + \beta^{T-1} F(k_T, k_{T+1})}_{T+1 \text{ terms}} \mid \begin{array}{l} k_{T+1} \in \Gamma(k_T) \\ k_2 \in \Gamma(k_1) \end{array} \right\} \mid k_1 \in \Gamma(k_0) \right\} \\
 &\quad \neq V_0(k_1) \text{ because the objective function has a different structure}
 \end{aligned}$$

denote it by $V_1(k_1)$

$$\Rightarrow V_0(k_0) = \max_{k_1 \in T(k_0)} \{F(k_0, k_1) + \beta V_1(k_1)\}$$

$$\boxed{V_t(k) = \max_{k' \in T(k)} \{F(k, k') + \beta V_{t+1}(k')\}, t=0 \dots T-1}$$

at T : $k' = 0 \Rightarrow V_T(k) = u(\underbrace{f(k)}_{\text{consume everything}})$

$\Rightarrow$ The sequence of $\{V_t(k)\}_{t=0}^T$ can be solved for "backwards".